

Characterization of the Molecule Release from Magnetic Nanoparticles

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Introduction

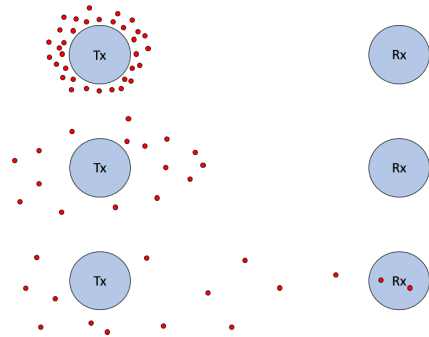


Introduction

- Nanotechnology refers to technologies where the size of the basic components are in the range of 1 nm to a few 100 nm.
- Health care applications have seen huge improvements due to recent technological developments, and have been able to develop powerful drugs for diseases that were previously untreatable.
- However, direct administration of these drugs result in a wide variety of side effects including fatigue, nausea, immune suppression, etc.
- Nanotechnology can help Medicine take the next big step by enabling highly targeted drug delivery (TDD) [1].

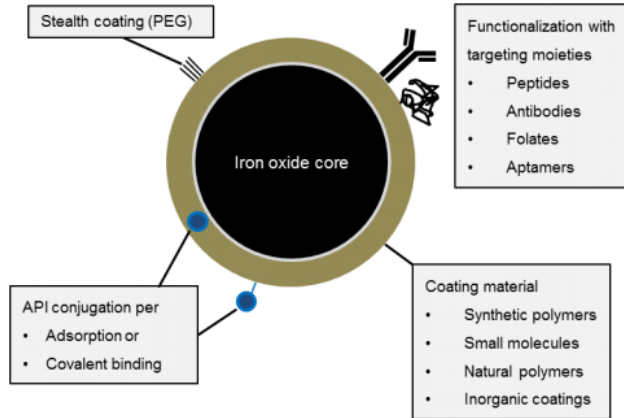
Introduction Contd.

- Guiding the drug containing nanoparticles to the target can significantly minimize the side effects.
- It can also help minimizing drug wastage and overall drug toxicity in the recipient, and improve the chances of a better outcome.
- Superparamagnetic Iron Oxide Nanoparticles (SPIONs) are one of such nanoparticles being developed for guided drug delivery.
- Guiding of the particles is achieved by exploiting super-paramagnetism of iron oxide (Fe_3O_4)



Schematic figure showing nanoparticle guided drug delivery

Introduction Contd.



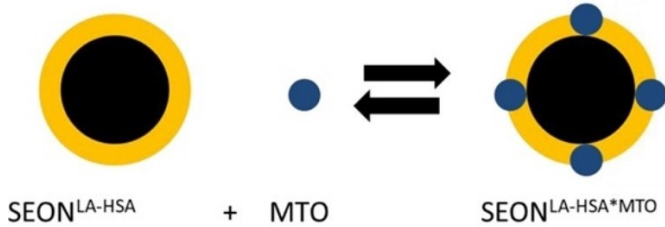
Functionalization of SPIONs from [1]

Design & structure



Manufacturing process

- Section for Experimental Oncology and Nanomedicine (SEON) team from Erlangen has developed $SEON^{LA-HSA*MTO}$ for TDD in humans. [2, 3, 4, 5]
- First, iron salts are dissolved in a specific solution to form an iron oxide precipitate.
- Then, lauric acid (LA) is added to the precipitate to form a colloid.
- For the colloid to maintain stability in fluids like blood, an extra protein shell of Human Serum Albumin (HSA) is added.
- The desired drug (MTO) is then added to this stable colloid, which gets adsorbed onto the HSA layer.
- This solution can be intravenously injected, and steered to the target site by a magnetic field gradient.



Drug binding to $SEON^{LA-HSA}$ from [6]

Drug binding



Binding site

- The desired drug predominantly binds to the HSA layer in the nanoparticle, which is why it is necessary to understand how and where the binding occurs, to estimate the release kinetics.
- Computational simulations in [7], and Spectroscopic and molecular modeling in [8] studied the binding mechanism of the drug in free HSA solution, and found multiple sites where the drug could potentially bind.
- It was discovered that two of the total available sites are the most important, and these lead to the most stable binding.
- Bond strength was observed to be not the same for both of the sites, which means that release from each of the sites might be different in nature.
- As there are no studies on the binding of the drug with the HSA layer on the nanoparticles yet, it is assumed that both of these sites are fully available in SPIONs.

Binding site contd.

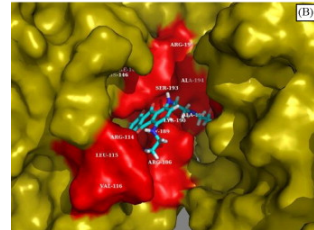
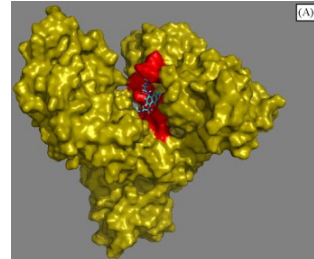
- However, if future spectroscopic results reveal otherwise, the model should be modified accordingly.
- Note that the solution containing the $SEON^{LA-HSA}$ particles could have free HSA molecules in it, which could be potential binding sites for the drug.
- If the drug binds to the free HSA molecules, this portion of the drug cannot be steered with external intervention.
- It is also observed that the drug bound to the free HSA will result in a uncontrolled burst release when introduced into the medium.
- Therefore, filtration of the excess HSA is necessary to minimize drug wastage.

Binding mechanism

- To measure the spontaneity of the binding reaction, Gibbs free energy (ΔG), and the underlying enthalpy (ΔH) and entropy (ΔS) were calculated in [8].
- This computations revealed that the ΔS contributes the most to ΔG due to the result of hydrophobic interactions between HSA and the drug.
- Although small, ΔH due to the electrostatic interactions and hydrogen-bonds cannot be neglected, as these bonds provide additional stability to the drug binding.
- Both of these interactions have to be taken into account while modeling the release pattern.



MTO molecule bound to HSA [7]



Polar residues due to MTO binding to HSA [8]

Drug release models



Factors affecting drug release and accumulation

Below are some factors listed, that can affect the release and accumulation pattern of the drug [9]:

- pH
- Temperature
- Nature of the bond between nanoparticle and the drug
- Diffusion rate of the drug, and presence of medium flow
- Presence of enzymes, catalysts, and inhibitors
- Drug distribution on the surface of nanoparticle
- Presence of competing targets

Drug release kinetics

- Unless the detailed mechanism of the reaction is known, the kinetics of a reaction are typically computed using model-fitting of experimental data.
- Common drug release kinetics include zero order, and first order release.
- The previous iteration of SEON particles exhibited a pseudo-zero-order release.
- However, there were some limitations with the previous measuring method, so we assume that both the orders are possible.
- Therefore, for the modeling of the release, we use a linear combination of zero and first order release. This is also inline with the observation that both the bonding sites have different bond strength.

Release model

- As per the above assumptions, we shall have a linear combination of zero order and first order.
- If the HSA particles in the solution are not filtered, then we might also need to accommodate a burst release term.
 - Zero order: $C_z(t) = \max\{P_z K_z t, P_z\}$
 - First order: $C_f(t) = P_f(1 - e^{-K_f t})$
 - Burst release: $C_b(t) = \frac{P_b}{1 + e^{-K_b t}} - P_0$
- Here, $P_z, P_f, \& P_b$ are the drug payloads, and $K_z, K_f, \& K_b$ are the reaction rates.
- The overall model would be:

$$C(t) = \max\{P_z K_z t, P_z\} + P_f(1 - e^{-K_f t}) + \frac{P_b}{1 + e^{-K_b t}} - P_0$$

Model computations

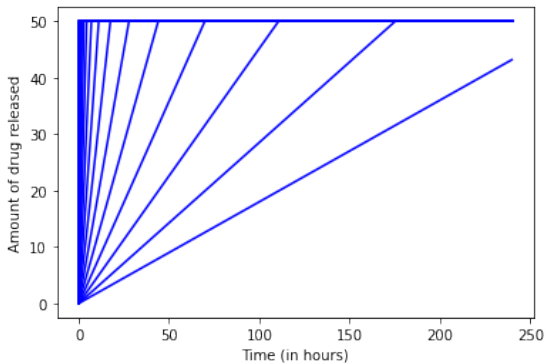


Model used for computations

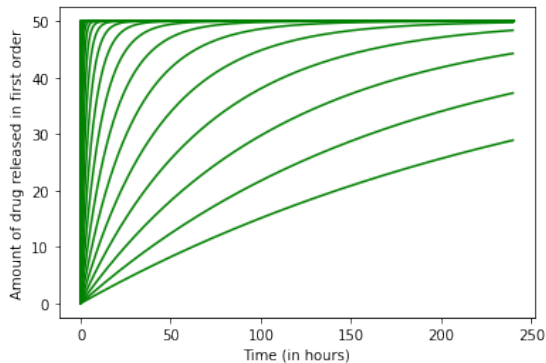
- In the new iteration of SPIONs, a filtration technique is being employed to minimize the drug content bound to free HSA.
- Therefore, we shall ignore the burst release component P_b in our calculations.
- In this case, the model shall be:

$$C(t) = \max\{P_z K_z t, P_z\} + P_f(1 - e^{-K_f t}) \quad (1)$$

Individual release orders



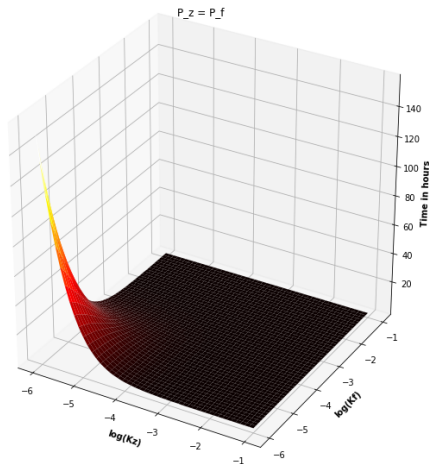
Zero order with increasing K_z



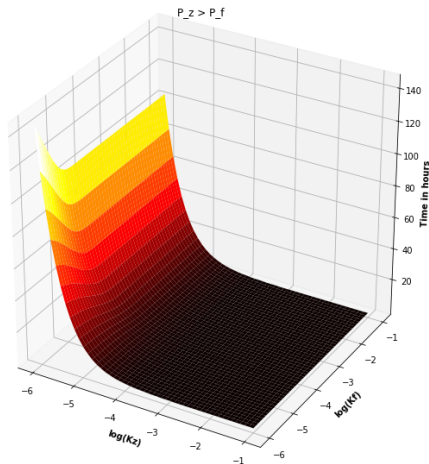
First order with increasing K_f

Time computations

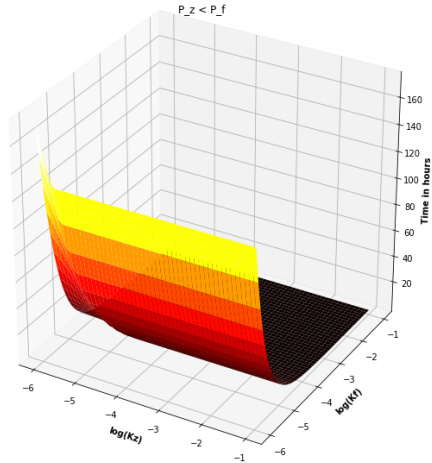
- Typically, we are interested in time taken for a certain amount of drug to be released.
- We can compute this time in three methods: experiments, simulation, and analytic computations.
- As experimental data is not available at the moment, we shall first compute the time by simulating the release and finding the first instance where the drug exceeds a certain threshold.
- Next, we shall try computing the time analytically, by solving equation (1) for a particular threshold $C(t)$



Time taken for 50% drug release, $P_z = P_f$



Time taken for 50% drug release, $P_z > P_f$



Time taken for 50% drug release, $P_z < P_f$

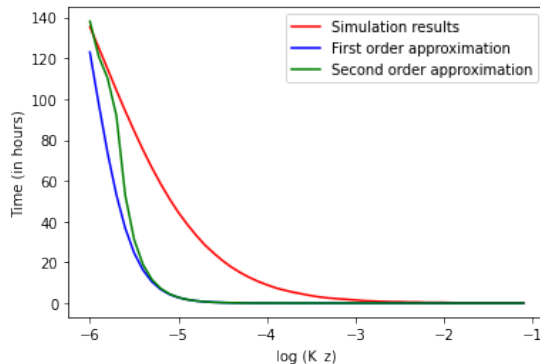
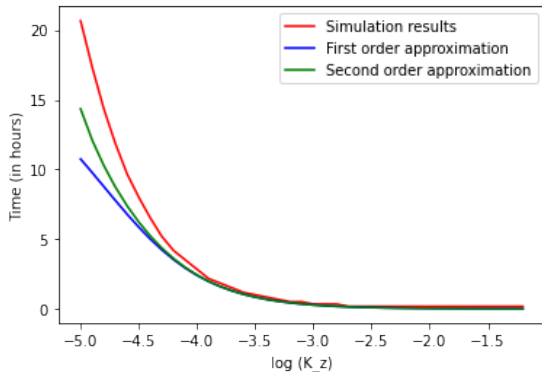
Analytical computations

- For analytical computation, we need to solve equation (1).
- However, it is not straightforward, as it has both polynomial and exponential terms.
- Therefore, we consider a Taylor series approximation of e^x for the computations.
- In this presentation, we shall calculate the time using the first and second order approximations of the exponential term.

Analytical computations

- Calculating the time by taking the first order approximation in equation (1) gives,
 - $t = \frac{C(t)}{P_z k_z + P_f K_f}$
- By using second order approximation and solving the quadratic equation, we get,
 - $t = \frac{P_z k_z + P_f k_f \pm \sqrt{(P_z k_z + P_f k_f)^2 - 2P_f k_f^2 C(t)}}{P_f K_f^2}$

Below figures show the difference between analytical and simulation models for different K_f



Limitations



Limitations of the model

- As we do not have the exact data on the order of drug release, two of the most common release rates were combined to build the model. However, if the drug exhibits an uncommon release behavior, a polynomial or Weibull model needs to be used.
- This model assumes that the release of drug particles through zero order and first order are independent of each other. This might not be true, if the re-adsorption rate of the drug is high.
- For simplicity of analytical computations, only until second order approximation of Taylor series for e^x was used. This is not accurate enough if the x becomes large.

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References IV

Thank You for Your Attention!

Questions?